INTRODUCTION TO HUMAN ANATOMY

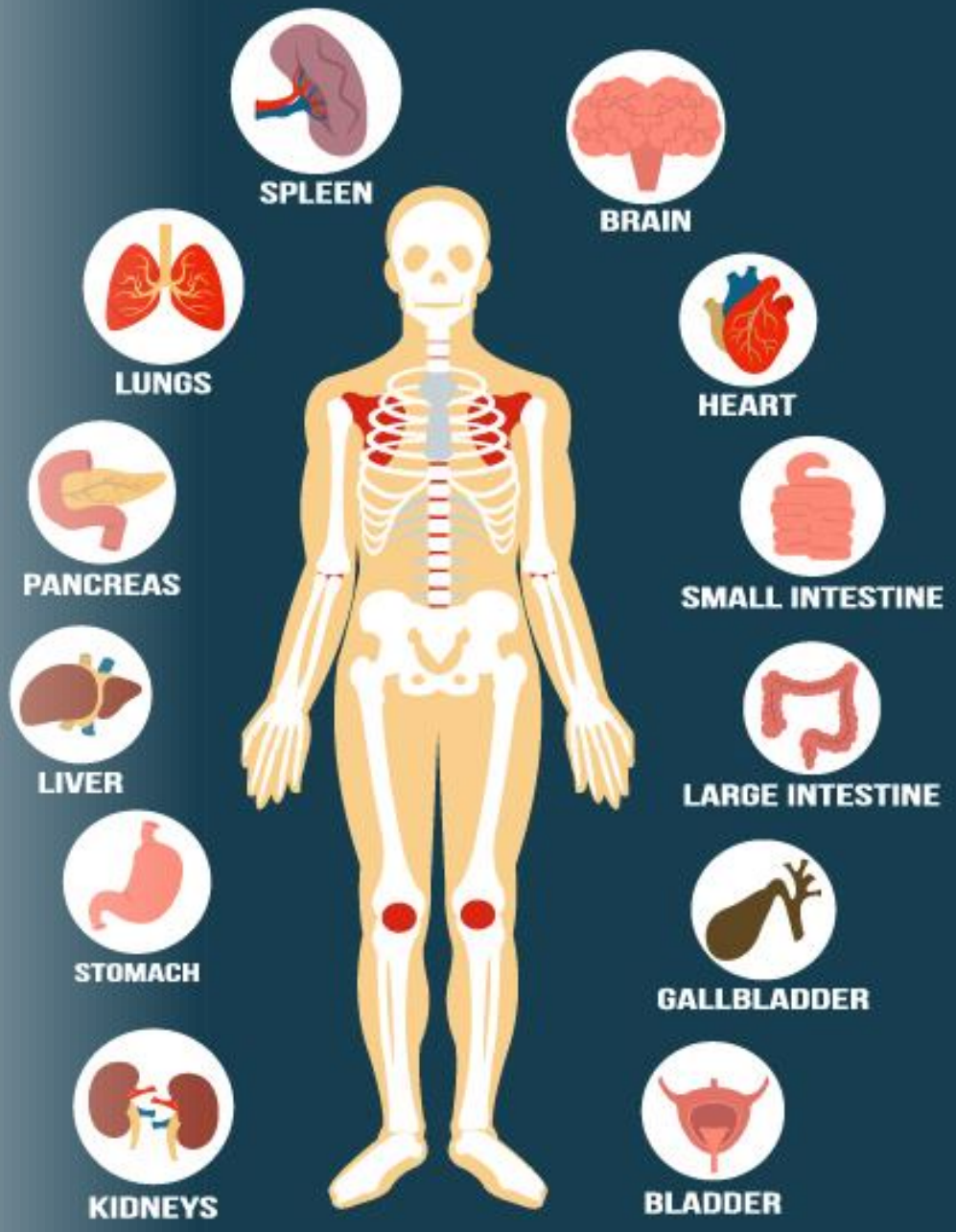
BY

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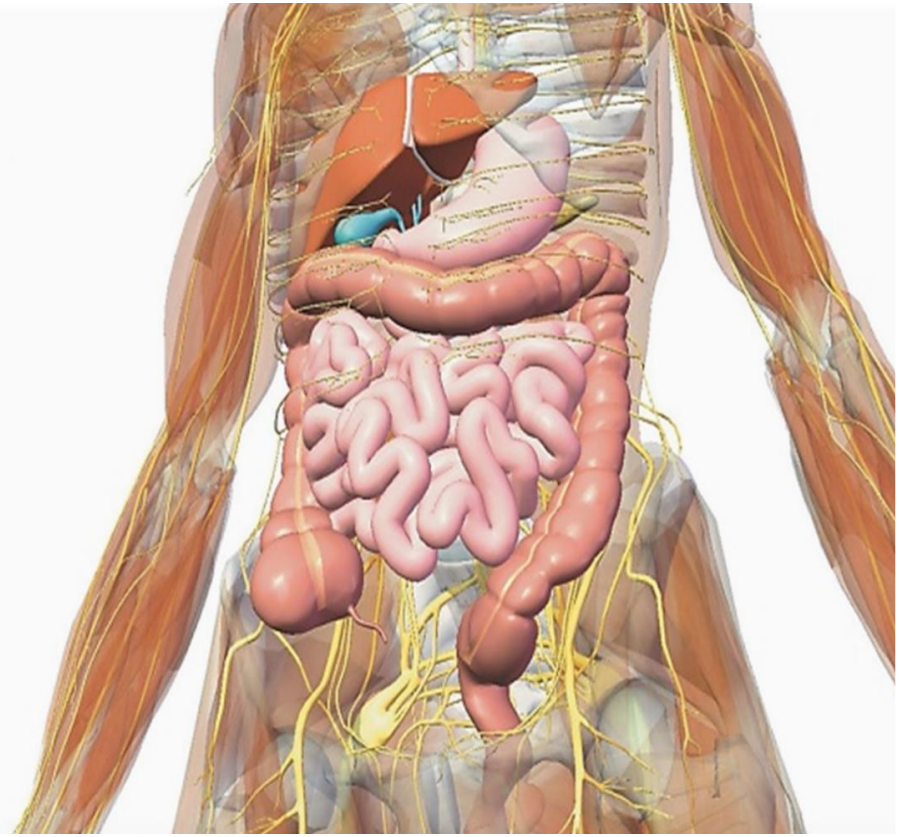
FACULTY OF MEDICINE

MENOUFIA UNIVERSITY



What is Anatomy ?

- Anatomy is the science which deals with the **structure** of the body and the **relationship** of its parts to each other.



Subdivisions of Anatomy

1) Macroscopic anatomy (Gross anatomy, Cadaveric anatomy):

- Is the study of dead body in dissecting room by naked eye.
- Can be done by two methods:
  - Systemic anatomy: the body is studied system by system. e.g. cardiovascular system.
  - Regional anatomy: the body is studied in regions e.g. upper limb, lower limb, abdomen, thorax.

2) Microscopic anatomy (Histology):

- It is the study of body structure by means of a microscope.

3) Embryology:

- Is the study of the changes that occur to fertilized ova before birth.

4) Surface anatomy:

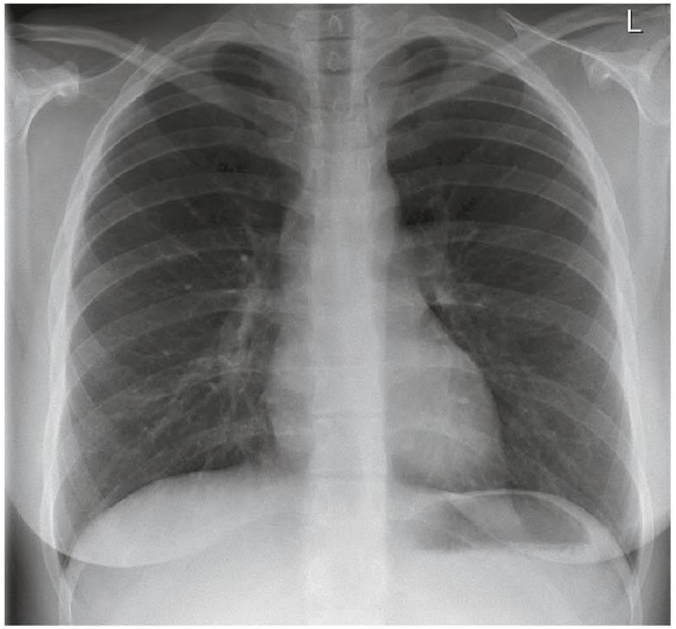
- Is the study of the deeper parts of the body in relation to the skin surface.

5) Radiographic anatomy and endoscopy:

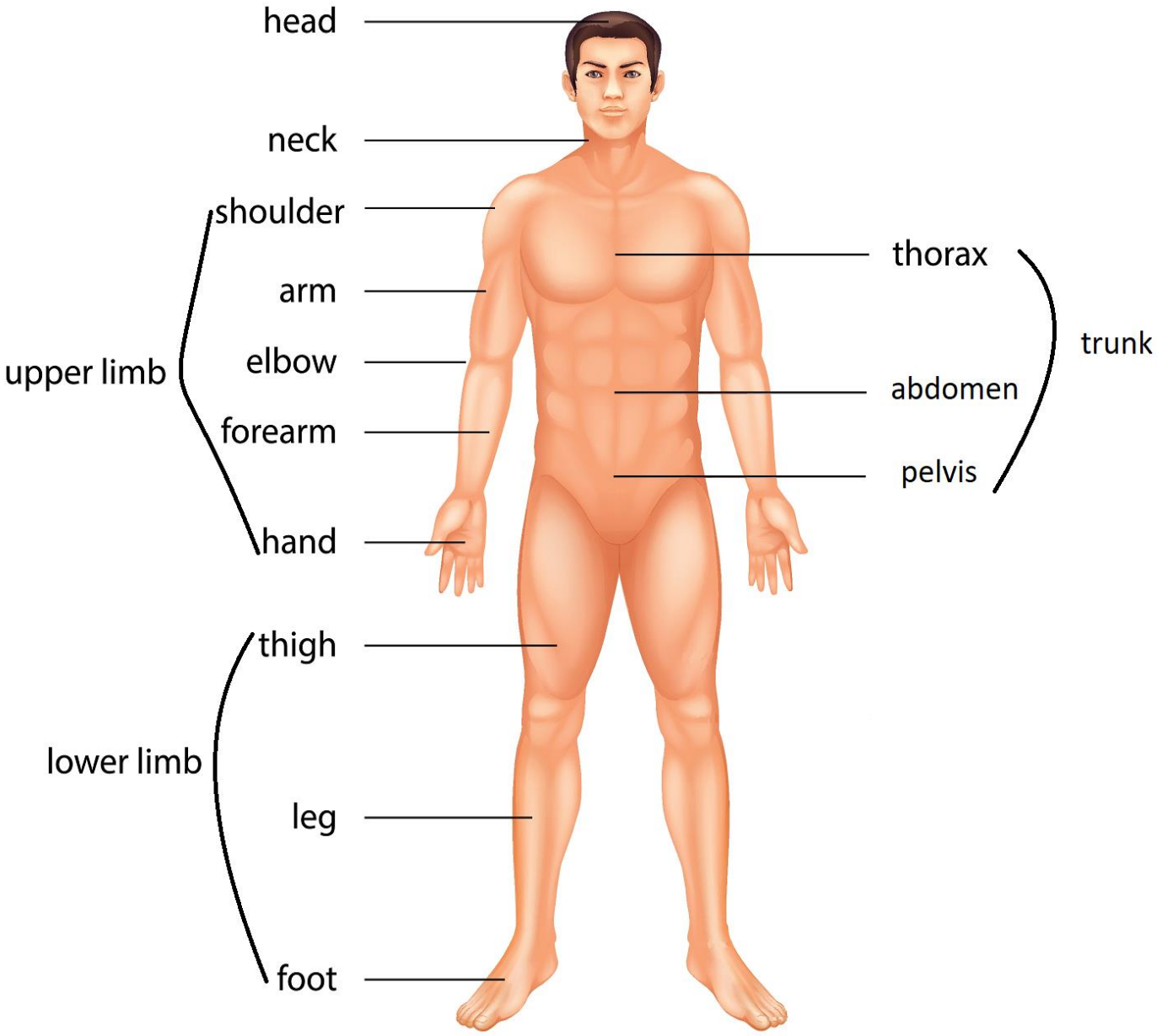
- Is the study of the invisible parts or organs by means of radiography, ultrasound and computerized tomographic (CT) scans and MRI scans and endoscope.

6) Applied anatomy:

- Is the application of the anatomical knowledge to the medical and surgical practice.

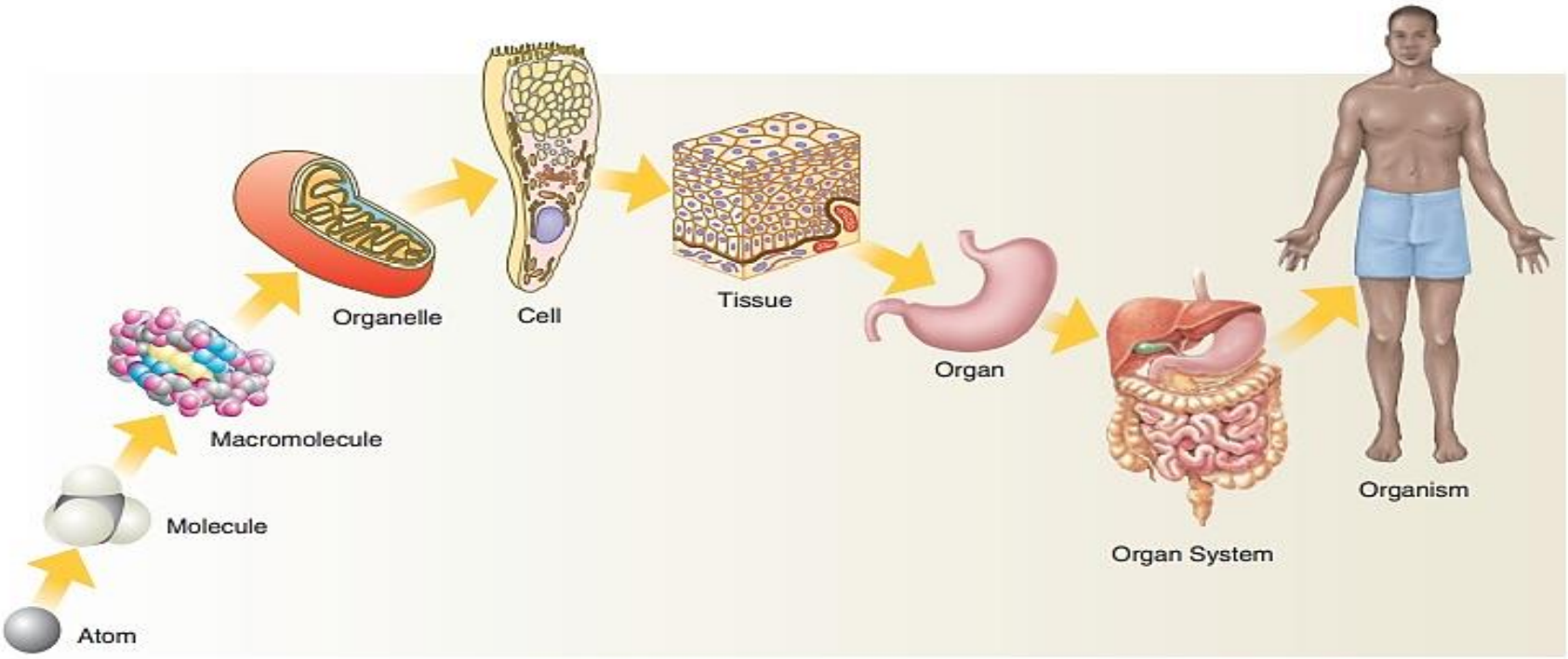


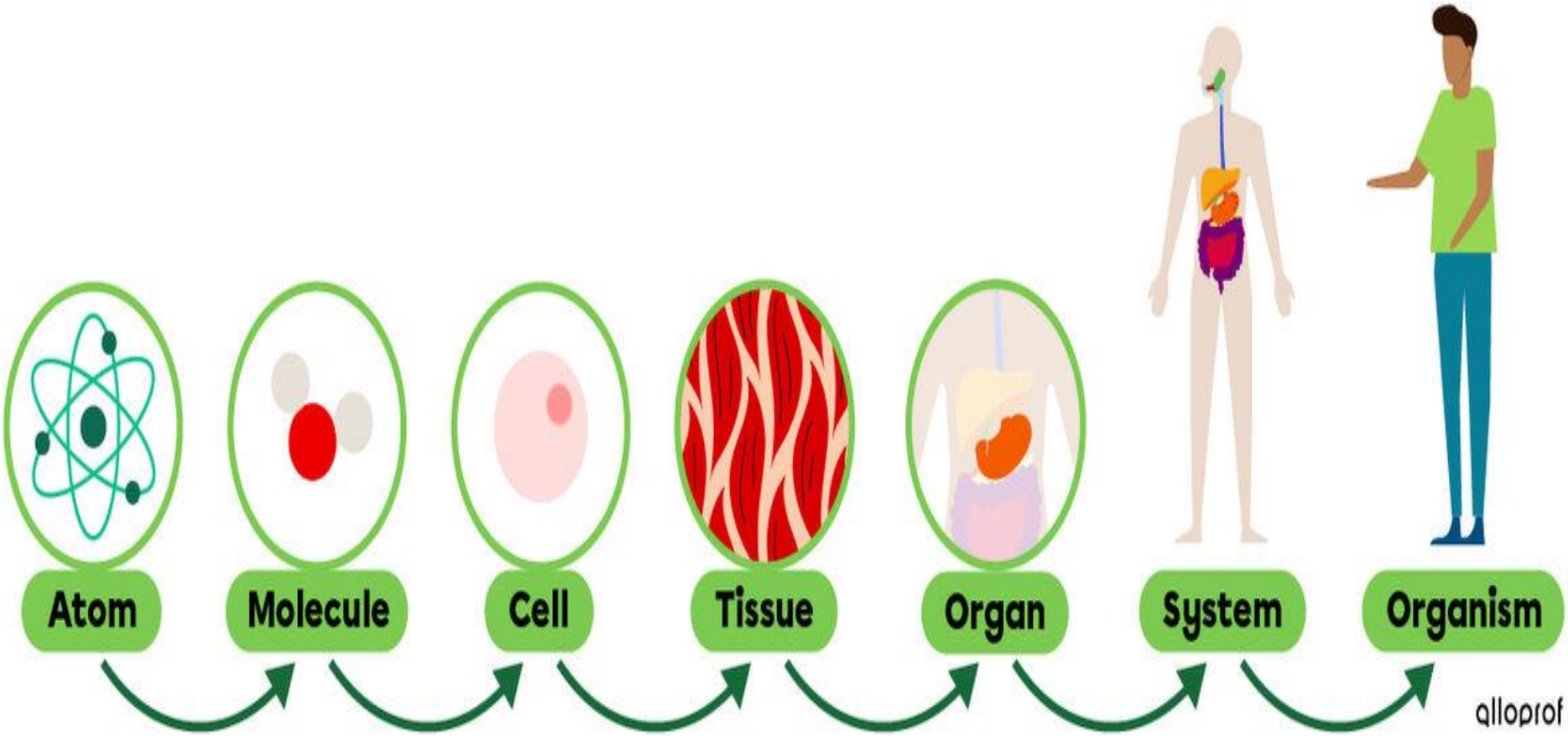
Body Regions



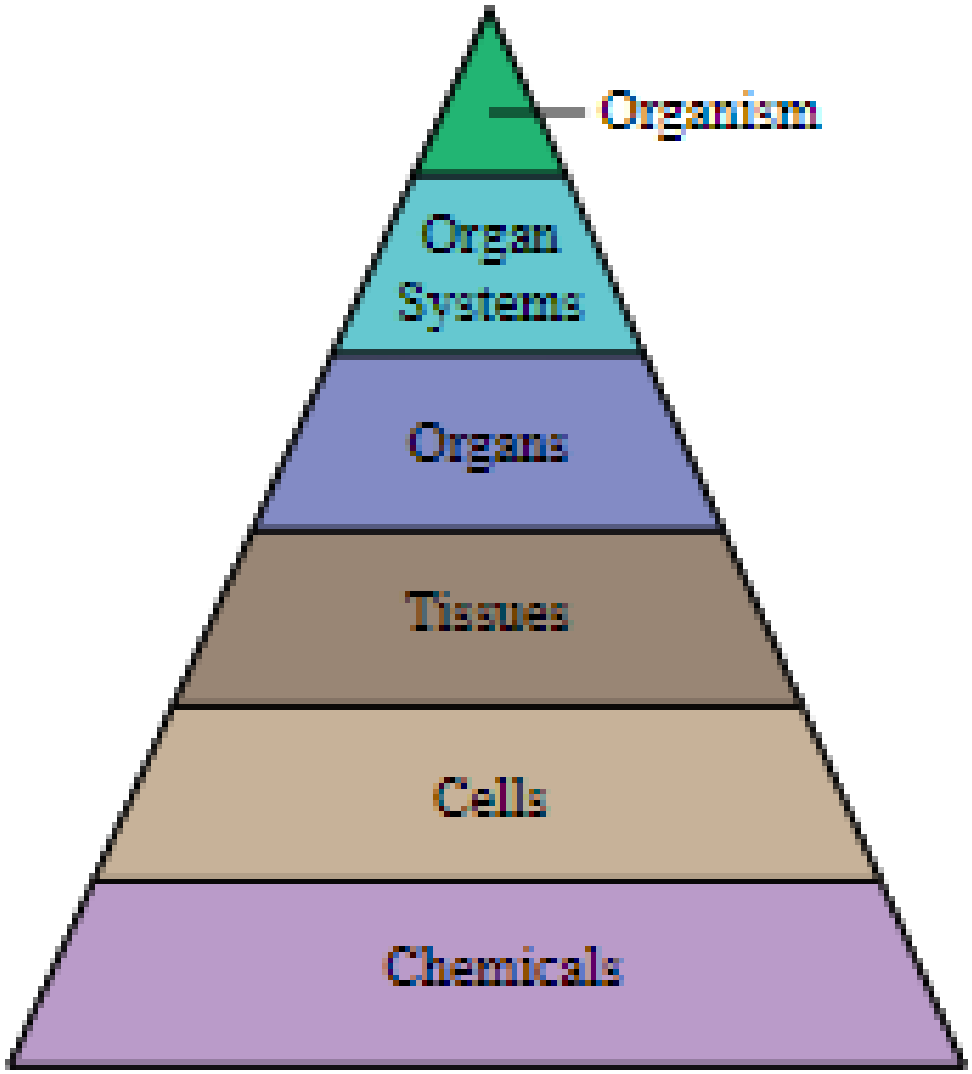
STRUCTURAL ORGANIZATION OF HUMAN BODY:

- **Chemical level:** Atoms→ Molecules →Organelles →the cell
- **Cellular level:** Cells, are the basic living units.
- **Tissues**: groups of cells that work together to perform a particular function, e.g. epithelial tissue, muscular tissue and nervous tissue.
- **Organs**: are made of different types of tissues, e.g. heart, kidney, lung and skin.
- **Organ system**: groups of organs function work together to perform a specific function.
- **The organism**: all the body systems functioning together to make up an organism which is the highest level of organization of the living human being.




Atom
Molecule
Cell
Tissue
Organ
System
Organism
alloprof

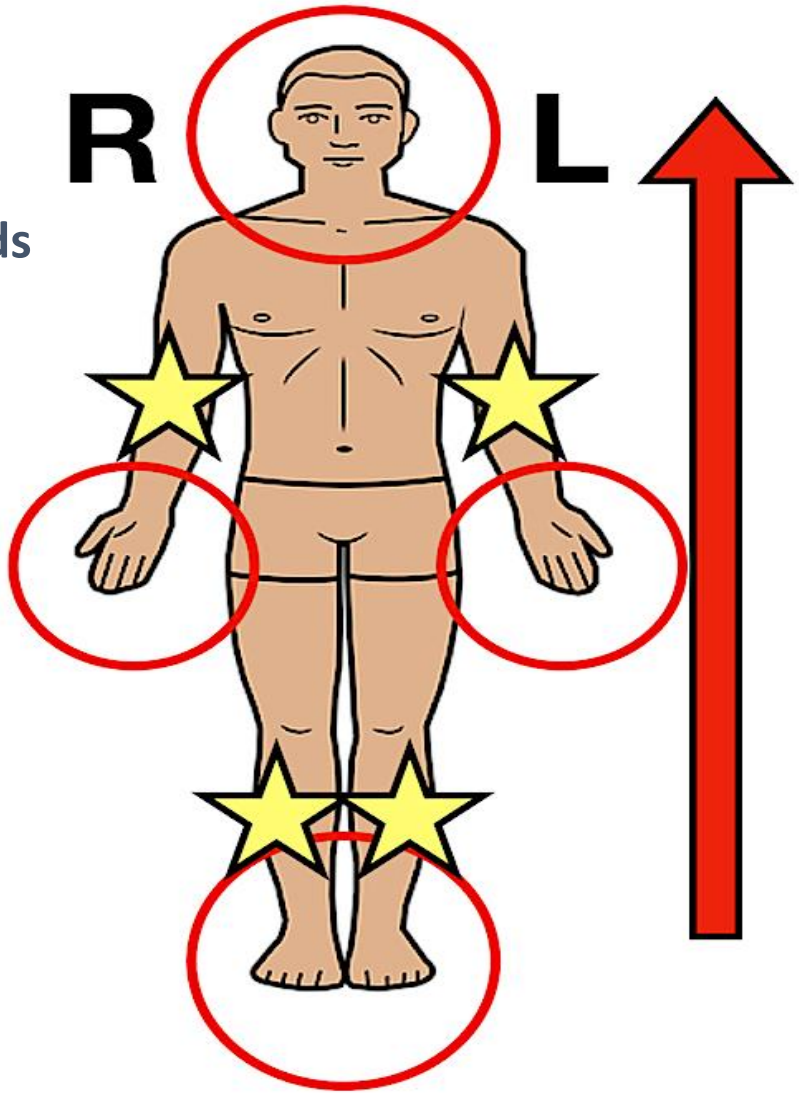
Structural Organization of the Body



The Anatomical position:

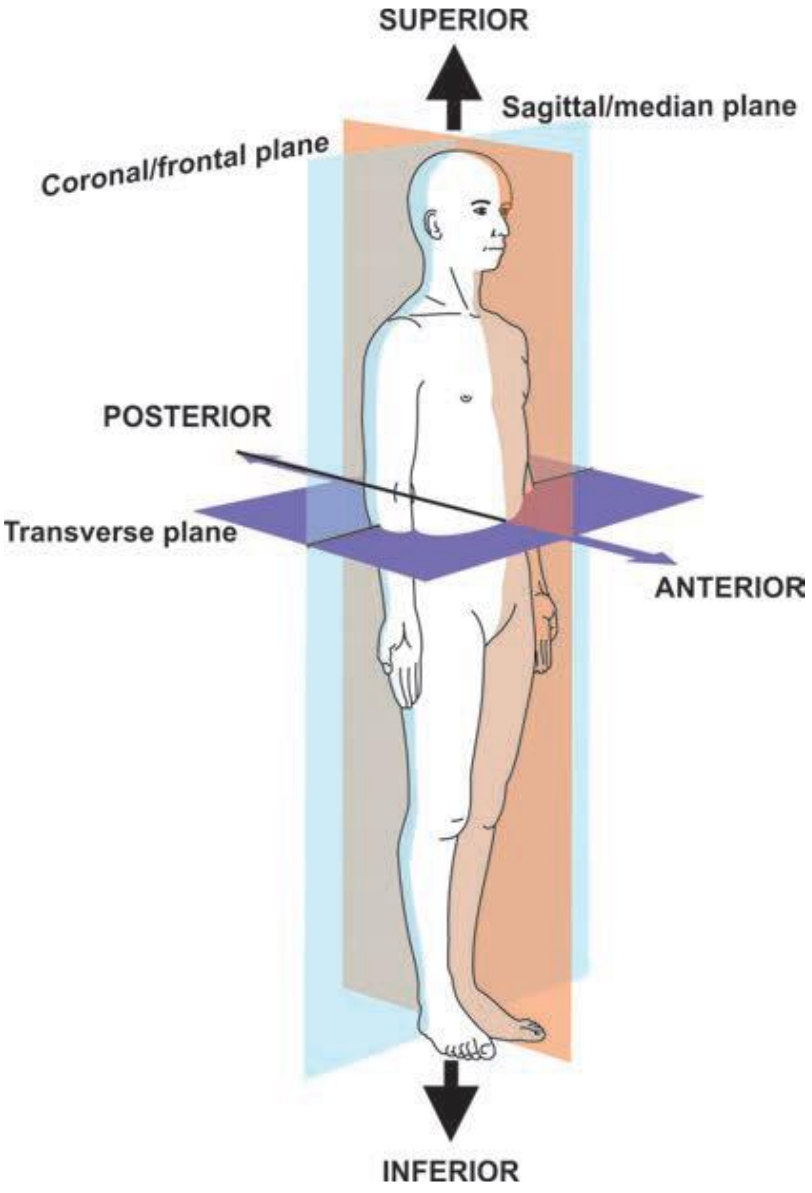
- The standard reference position of the body used to describe the location and the relationship of anatomical structures.

- ✓ Standing erect with his eyes looking forwards
- ✓ Arms resting by the side of the body and palms are facing forwards
- ✓ Feet flat on the floor



The Anatomical planes:

- ✓ These are imaginary planes that pass through the body to understand relations of internal structures (organs inside the body).
- ✓ Individual organs can also be studied by using similar planes of sectioning.



Anatomical planes:

1) Vertical

- The midsagittal plane or median plane.
- The sagittal planes.
- The coronal or frontal planes.

2) Horizontal

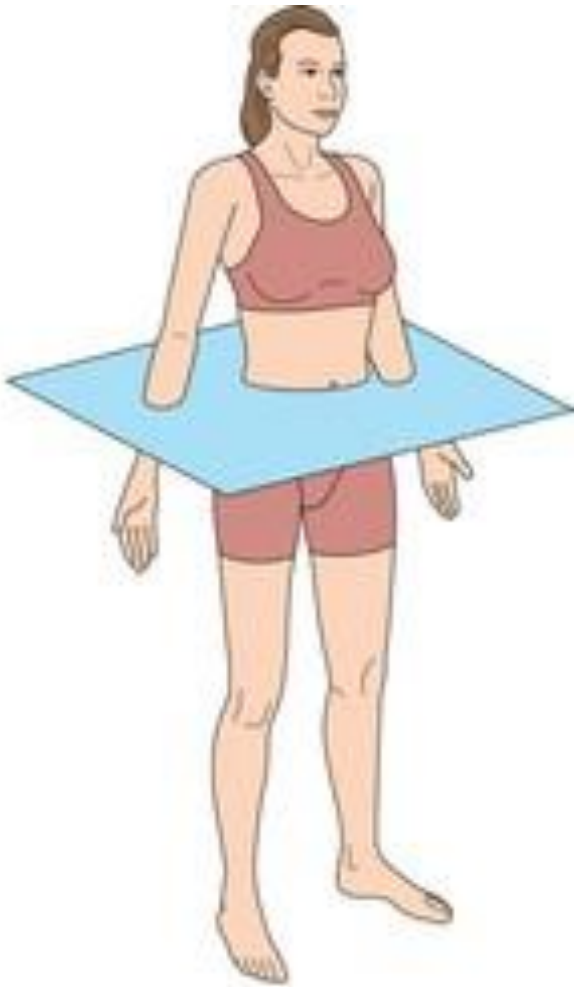
- The transverse or horizontal planes.



Sagittal plane



Frontal/coronal plane

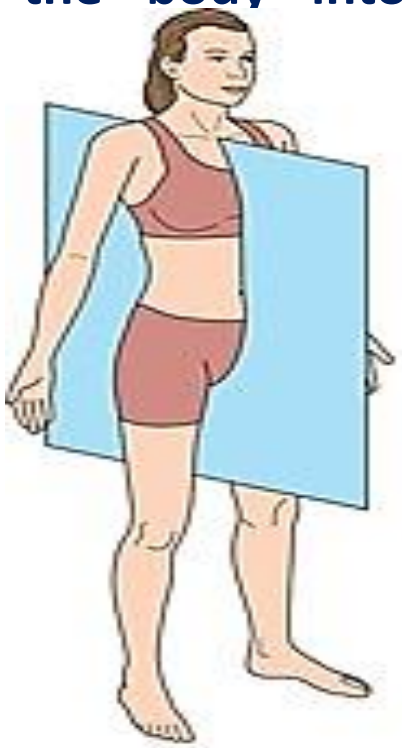


Transverse plane

The midsagittal plane or median plane

- The vertical plane passing longitudinally through the body dividing the body into right and left halves.

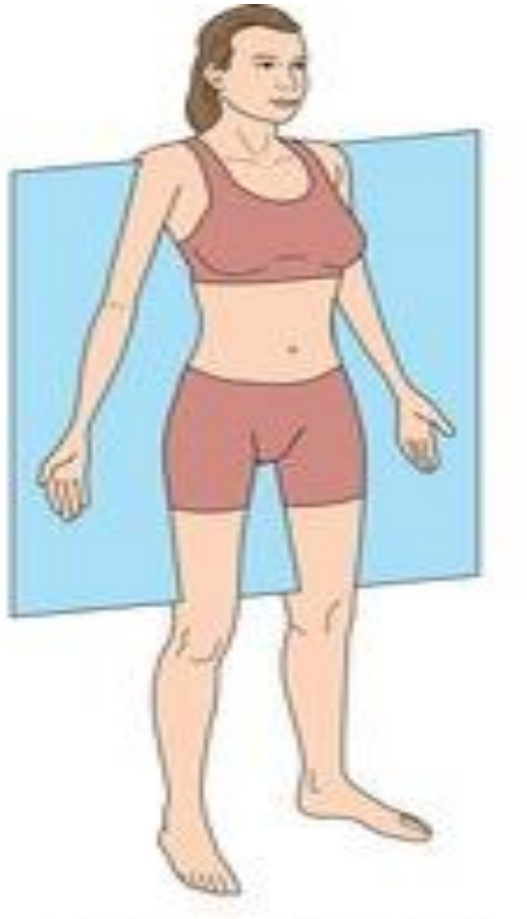
- **Sagittal planes** are vertical planes passing through the body parallel to the median plane. dividing the body into unequal longitudinal parts.



Sagittal plane

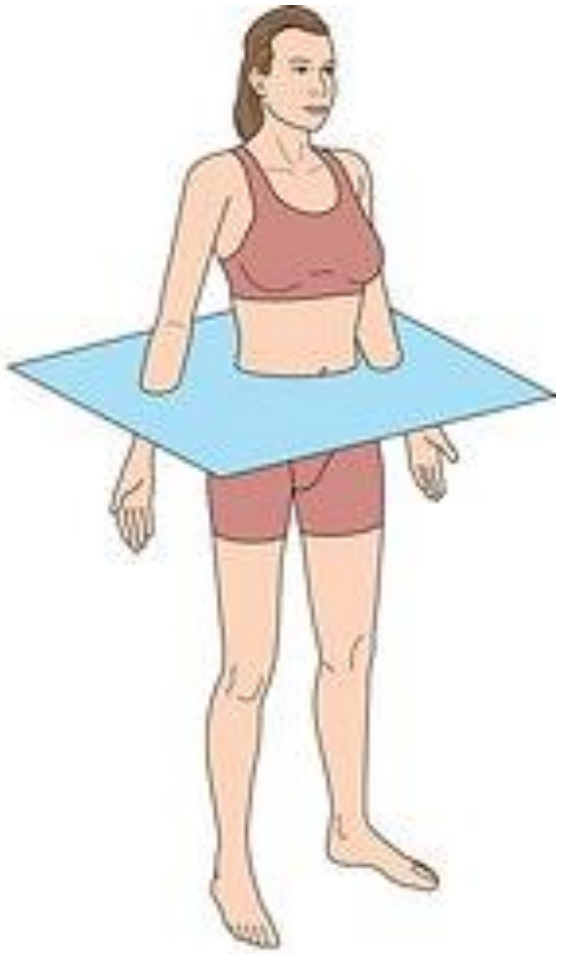
The coronal or frontal planes

- Vertical planes passing through the body at right angles to the median plane, dividing the body into anterior (front) and posterior (back) parts.

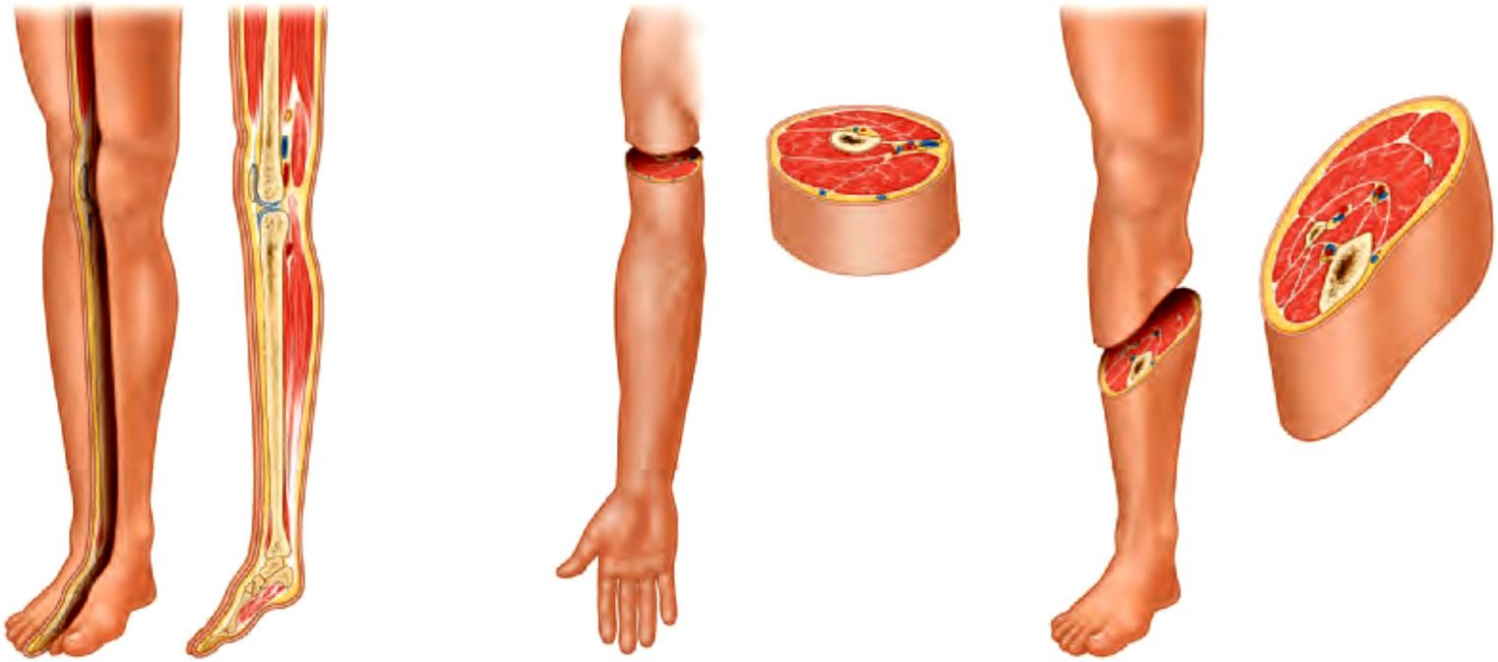


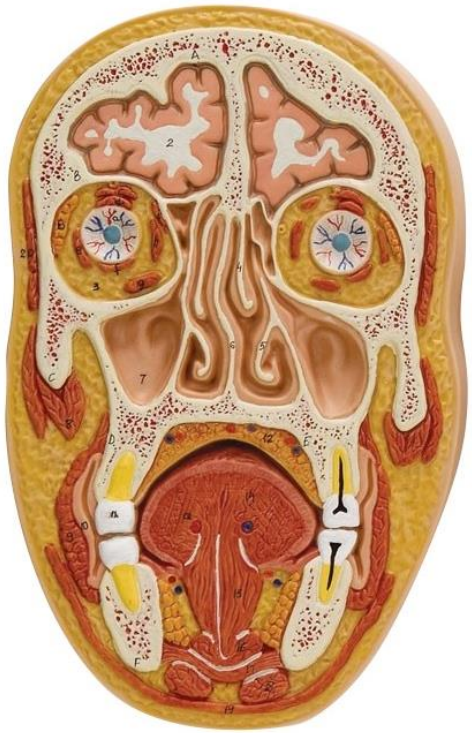
The transverse or horizontal planes

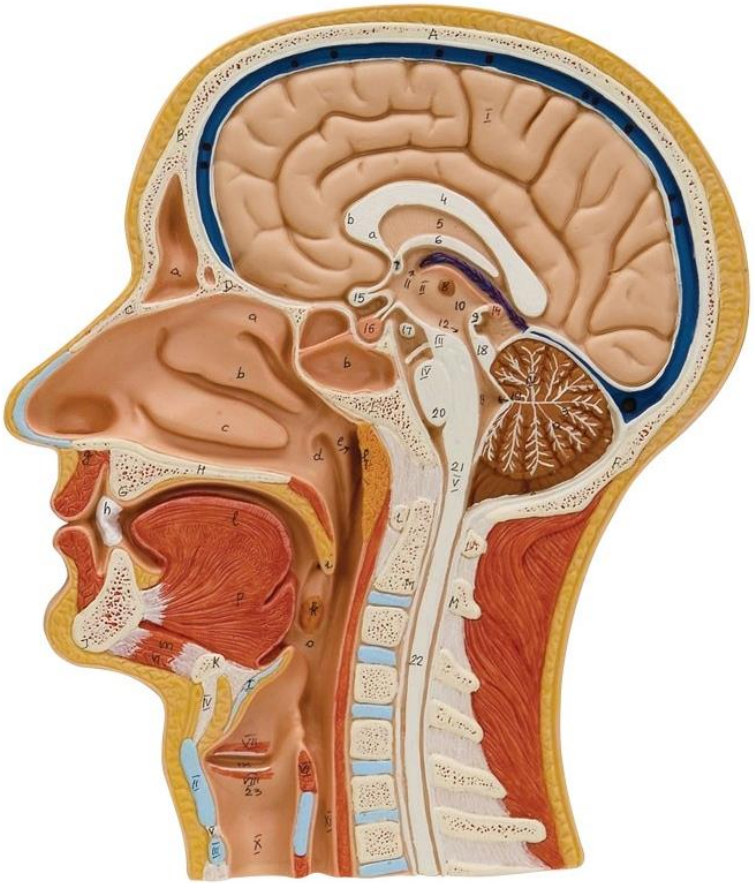
- **Horizontal planes passing through the body at right angles to the median and frontal planes, dividing the body into superior (upper) and inferior (lower) parts.**

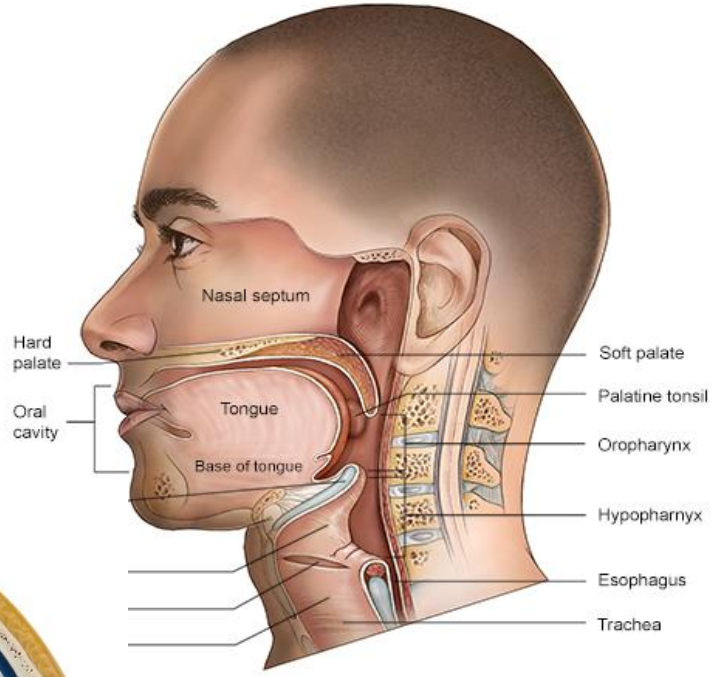


Transverse plane








Nasal septum
Hard palate
Oral cavity
Tongue
Base of tongue
Soft palate
Palatine tonsil
Oropharynx
Hypopharnyx
Esophagus
Trachea

Anatomical terms:

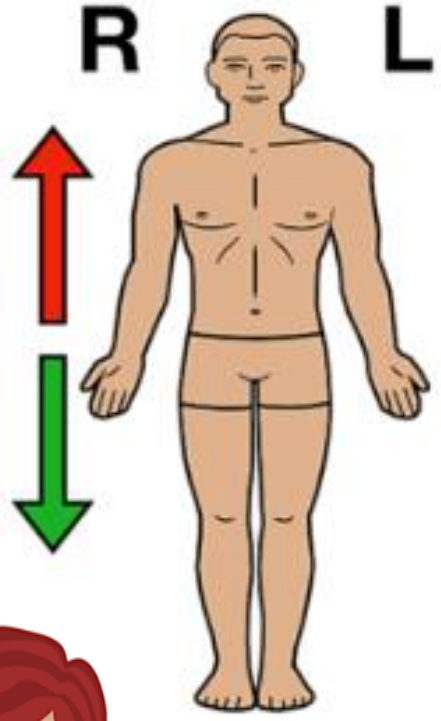
1) Terms related to position

2) Terms related to movement

Terms of position

Superior / Inferior

- ✓ Superior : nearer to the head.
- ✓ Inferior: away from the head and nearer to the feet.



Eyes are superior to the Nose

Nose is superior to the Mouth

Mouth is inferior to the Nose

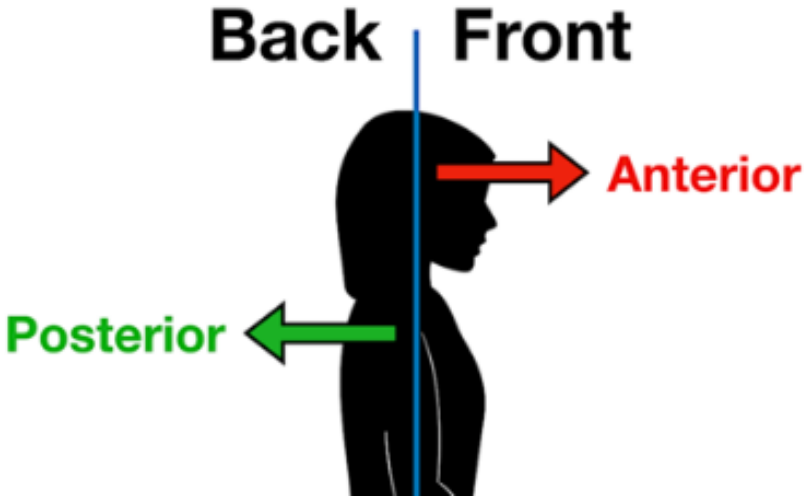
Nose is inferior to the Eyes



Terms of position

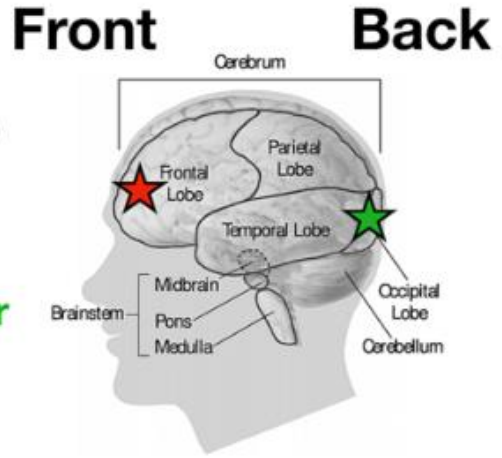
Anterior / Posterior

- ✓ Anterior (ventral): Nearer to or at the front of the body.
- ✓ Posterior (dorsal): Nearer to or at the back of the body.



The Frontal Lobe is anterior to the Occipital Lobe

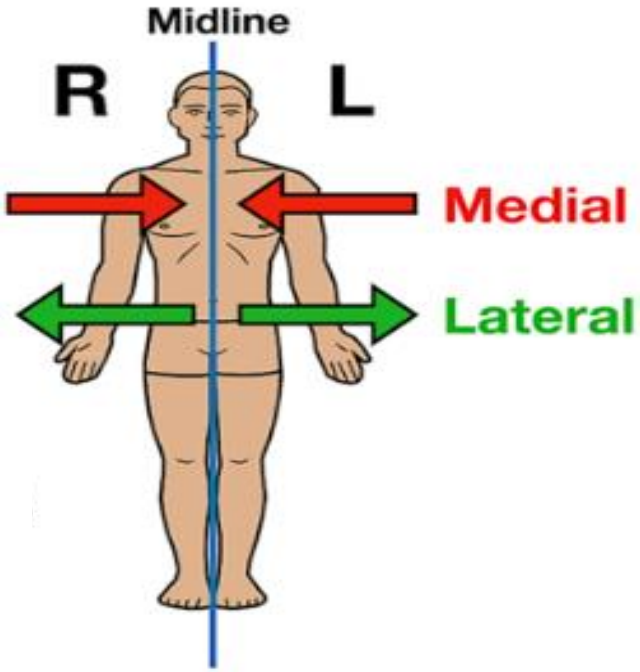
The Occipital Lobe is posterior to the Frontal Lobe



Terms of position

Medial / Lateral

- ✓ Medial: nearer to the median plane of the body.
- ✓ Lateral: away from the median plane of the body.



Lungs are lateral to the Heart

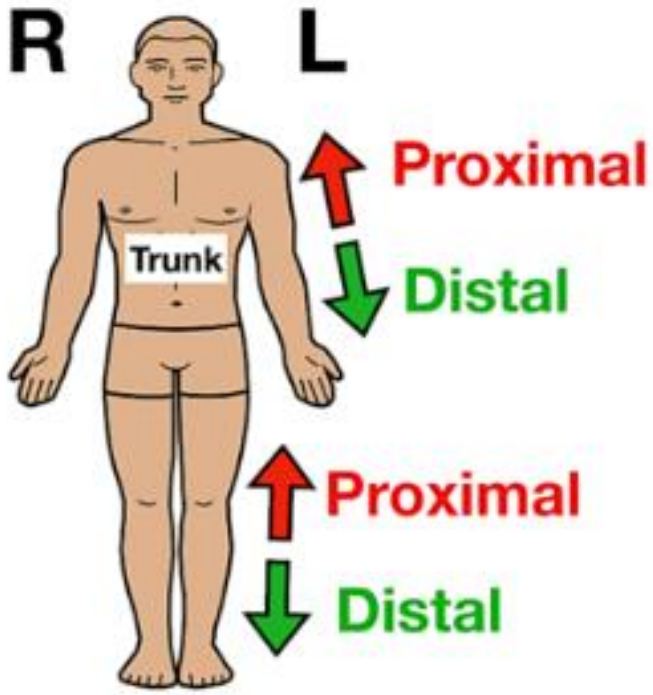
Heart is medial to the Lungs



Terms of position

Proximal / Distal

- ✓ Proximal: Nearer to the attachment of a limb to the trunk.
- ✓ Distal: away from the attachment of a limb to the trunk.

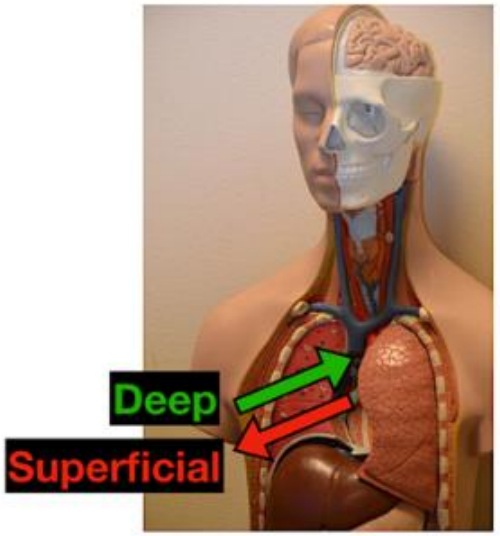


Wrist is proximal to the Hand
Elbow is proximal to the Wrist
Shoulder is proximal to the Elbow

Elbow is distal to the Shoulder
Wrist is distal to the Elbow
Hand is distal to the Wrist



Terms of position



Superficial / Deep

- ✓ Superficial: nearer to the surface of the body.
- ✓ Deep: away from the surface of the body.

Skin is superficial to the Ribs
Ribs are superficial to the Lungs

Lungs are deep to the Ribs
Ribs are deep to the Skin

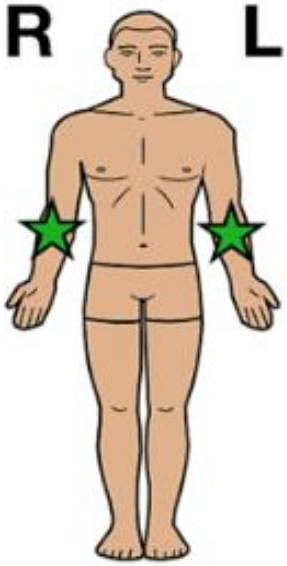


Terms of position

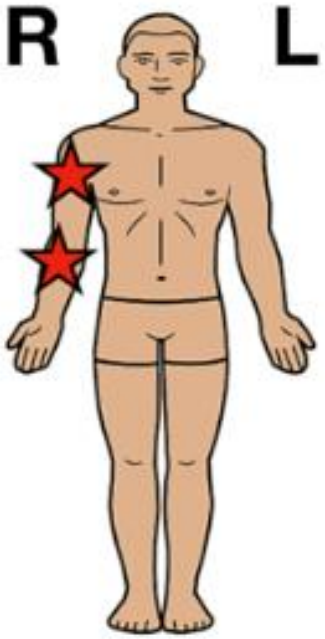
Ipsilateral / Contralateral

- ✓ Ipsilateral: occurring on the same side of the body.
- ✓ Contralateral: occurring on the opposite sides of the body.

Contralateral



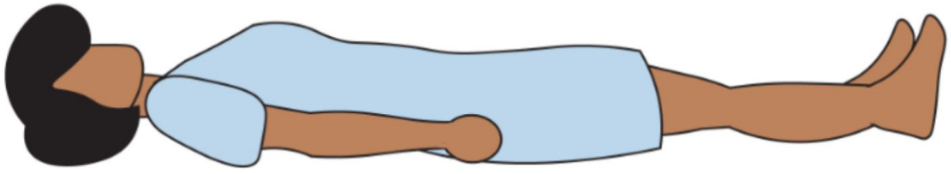
Ipsilateral



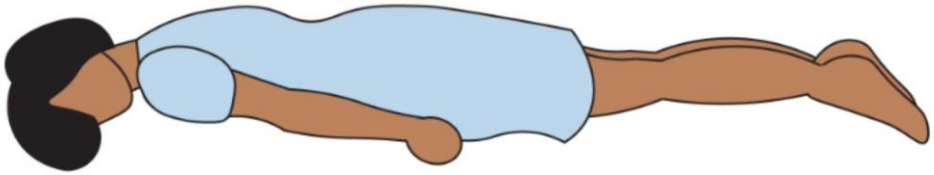
Terms of position

Supine / Prone position

- ✓ Supine position: the body is lying on the back.
- ✓ Prone position: the body is lying face down.



Supine



Prone


Thank
You